Alpha Diversity

Nitin Bayal

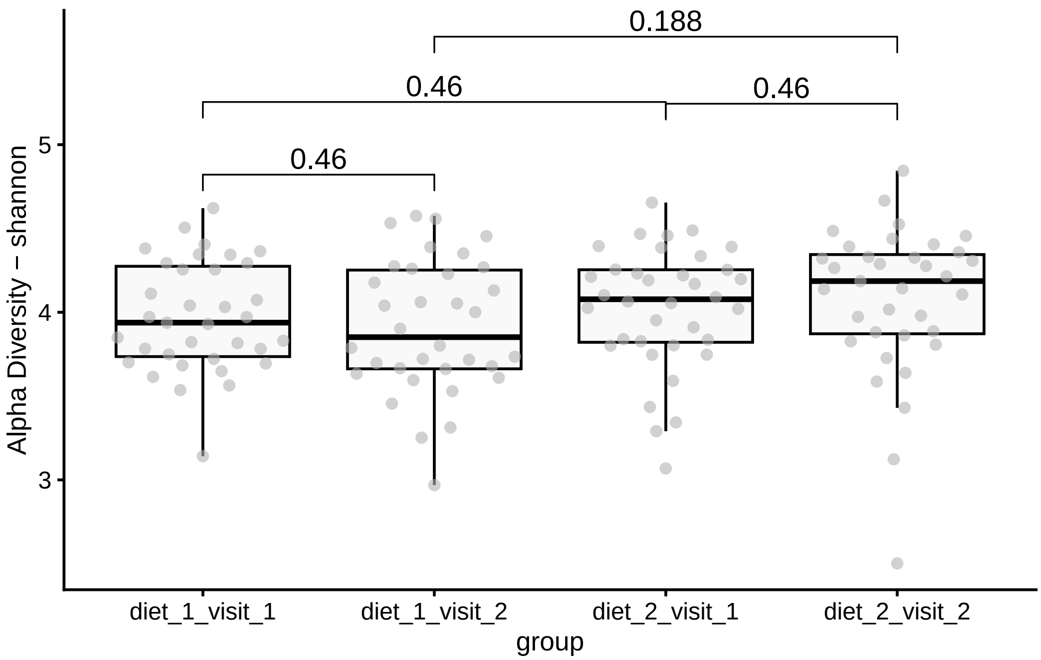
Introduction

This document reports on a microbiota project results of alpha diversity across different diets and visits.

Group-wise comparisons

- **Index:** shannon
- **Group variable:** group
- **Multiple testing correction:** fdr

Visualization



Characteristic	after			before		
	Beta	95% CI[1]	p-value	Beta	95% CI[1]	p-value
(Intercept)	3.5	2.9, 4.1	<0.001	3.6	3.1, 4.0	<0.001
meal_group			0.3			0.3
1	—	—		—	—	
2	0.28	-0.02, 0.58	0.065	0.20	-0.03, 0.44	0.085
3	0.20	-0.09, 0.50	0.2	0.11	-0.12, 0.35	0.3
4	0.10	-0.20, 0.39	0.5	0.17	-0.06, 0.40	0.2
age	0.01	0.00, 0.02	0.2	0.01	0.00, 0.01	0.2

[1]CI = Confidence Interval

Paired test of before and after	Beta	95% CI[1]	p-value
(Intercept)	3.5	3.2, 3.9	<0.001
meal_group			0.11
1	—	—	
2	0.23	0.04, 0.42	0.017
3	0.16	-0.03, 0.34	0.094
4	0.13	-0.05, 0.31	0.2
age	0.01	0.00, 0.01	0.039

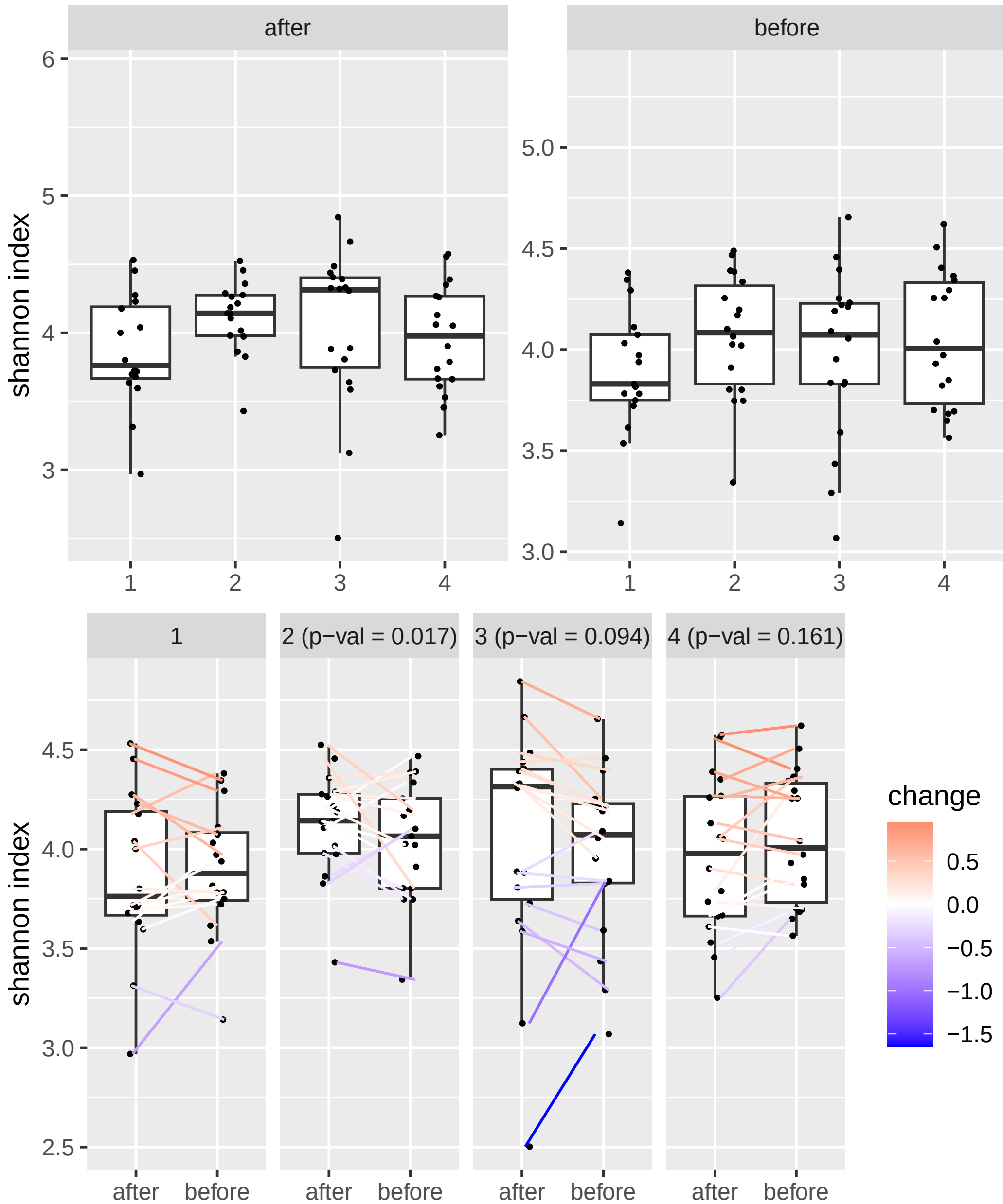
[1]CI = Confidence Interval

Table of the top findings

Table 1: Table of alpha diversity comparisons

group1	group2	mean1	mean2	logFC	p.adj
diet_1_visit_1	diet_1_visit_2	3.97	3.91	-0.02	0.460
diet_2_visit_1	diet_2_visit_2	4.02	4.08	0.02	0.460
diet_1_visit_1	diet_2_visit_1	3.97	4.02	0.02	0.460
diet_1_visit_2	diet_2_visit_2	3.91	4.08	0.06	0.188

Alternate analyses



Data Summary

```
class: TreeSummarizedExperiment
dim: 1700 140
```

```
metadata(0):
assays(2): counts relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AK1371 ... TS2358 VK2336
colData names(13): sample id ... shannon group
reducedDimNames(0):
mainExpName: NULL
altExpNames(10): PrevalentGenus PrevalentSpecies ... species strain
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL
```

Version info

```
R version 4.4.2 (2024-10-31)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 24.04.1 LTS

Matrix products: default
BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0

locale:
 [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
 [3] LC_TIME=en_US.UTF-8        LC_COLLATE=en_US.UTF-8
 [5] LC_MONETARY=en_US.UTF-8    LC_MESSAGES=en_US.UTF-8
 [7] LC_PAPER=en_US.UTF-8       LC_NAME=C
 [9] LC_ADDRESS=C               LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C

time zone: Europe/Helsinki
tzcode source: system (glibc)

attached base packages:
[1] stats4    stats     graphics  grDevices utils     datasets  methods
[8] base

other attached packages:
 [1] kableExtra_1.4.0               vegan_2.6-8
 [3] lattice_0.22-6                 permute_0.9-7
```

```
 [5] lubridate_1.9.3                  forcats_1.0.0
 [7] stringr_1.5.1                    purrr_1.0.2
 [9] readr_2.1.5                      tibble_3.2.1
[11] tidyverse_2.0.0                  tidyr_1.3.1
[13] scater_1.33.4                    scuttle_1.15.4
[15] quarto_1.4.4                     patchwork_1.3.0
[17] parameters_0.24.0                multtest_2.61.0
[19] miaTime_0.1.21                   miaViz_1.15.4
[21] ggraph_2.2.1                     mia_1.15.6
[23] TreeSummarizedExperiment_2.13.0 Biostrings_2.73.2
[25] XVector_0.45.0                   SingleCellExperiment_1.27.2
[27] MultiAssayExperiment_1.31.5     SummarizedExperiment_1.35.3
[29] Biobase_2.65.1                   GenomicRanges_1.57.1
[31] GenomeInfoDb_1.41.2              IRanges_2.39.2
[33] S4Vectors_0.43.2                 BiocGenerics_0.51.3
[35] MatrixGenerics_1.17.0            matrixStats_1.4.1
[37] gtsummary_2.0.4                  ggsignif_0.6.4
[39] ggpubr_0.6.0                     ggplot2_3.5.1
[41] DT_0.33                          dplyr_1.1.4
[43] cardx_0.2.2                      ANCOMBC_2.8.0

loaded via a namespace (and not attached):
  [1] gld_2.6.6                   nnet_7.3-19
  [3] TH.data_1.1-2               vctrs_0.6.5
  [5] energy_1.7-12               digest_0.6.37
  [7] proxy_0.4-27                Exact_3.3
  [9] rbiom_1.0.3                 ggrepel_0.9.6
 [11] bayestestR_0.15.0           parallelly_1.38.0
 [13] mediation_4.5.0             MASS_7.3-61
 [15] reshape2_1.4.4              foreach_1.5.2
 [17] withr_3.0.2                 xfun_0.49
 [19] ggfun_0.1.6                 survival_3.7-0
 [21] doRNG_1.8.6                 memoise_2.0.1
 [23] commonmark_1.9.1            ggbeeswarm_0.7.2
 [25] emmeans_1.10.4              gmp_0.7-5
 [27] systemfonts_1.1.0           tidytree_0.4.6
 [29] zoo_1.8-12                  gtools_3.9.5
 [31] Formula_1.2-5               datawizard_0.13.0
 [33] httr_1.4.7                  rstatix_0.7.2
 [35] globals_0.16.3              ps_1.8.0
 [37] rstudioapi_0.17.1           UCSC.utils_1.1.0
 [39] generics_0.1.3              base64enc_0.1-3
 [41] processx_3.8.4              zlibbioc_1.51.1
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[123] gt_0.11.1                   cards_0.4.0
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[183] htmlwidgets_1.6.4          rlang_1.1.4
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